



ABG ANALYSIS SIMPLIFIED PART 1



NORMAL REFERENCE LEVEL

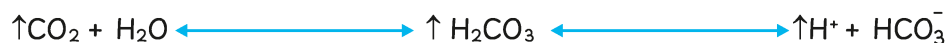
pH	7.35 - 7.45
PO ₂	75 - 100 mmHg
PCO ₂	35 - 45 mmHg
HCO ₃ ⁻	22 - 26 meq/L



RESPIRATORY ACIDOSIS

00:01:40

- Acute exacerbation chronic bronchitis



- Compensation (Respiratory Alkalosis): Kidney: PCT: HCO₃⁻ Reabsorption
- In acute compensation
 - 10 mm Hg ↑ in CO₂: 1 meq ↑ in HCO₃⁻
- In chronic compensation
 - 10 mm Hg ↑ in CO₂: 4 meq ↑ in HCO₃⁻
- ROME: Respiratory Opposite Metabolic Equivalence

pH	pCo ₂	HCO ₃	INTERPRETATION
↓	↓	↓	Metabolic Acidosis (Partially Compensated)
↑	↑	↑	Metabolic Alkalosis (Partially Compensated)
↓	↑	↑	Respiratory Acidosis (Partially Compensated)
↑	↓	↓	Respiratory Alkalosis (Partially Compensated)

CAUSES OF RESPIRATORY ACIDOSIS

Mnemonic: **DEPRESO**

- D**rugs: Morphine, Sedatives, Anaesthetics
- D**isease of Neuromuscular junction - Polio, Amyotrophic Lateral Sclerosis (motor neuron disease), Myasthenia Gravis (crisis), Muscular Dystrophy

- Edema and VP balance: ARDS (severe)
- Pleural effusions and barotrauma
- Respiratory Obstruction due to foreign body
- Spasm
 - Asthma (Life threatening)
 - Acute exacerbation of chronic bronchitis
- Obesity
 - Obstructive sleep apnea-hypopnea syndrome (OSAHS)
 - Obesity Hypoventilation Syndrome (Pickwinian Syndrome)

CLINICAL FEATURES OF RESPIRATORY ACIDOSIS

ACUTE

- Anxiety, confusion and hallucinations

CHRONIC

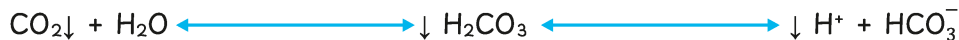
- ↑↑ daytime somnolence
- Asterixis / Flapping tremors
- ↑ ICP- papilledema

TREATMENT

- Elective Intubation and IPPV
- Aggressive T/t is not recommended → sudden CO₂ washout → loss of vasodilator effect of CO₂ on cerebral blood vessels → ↓ in Cerebral perfusion → resultant dizziness, seizures

RESPIRATORY ALKALOSIS

00:17:19



- Seen in Anxiety, Pneumonia, AHF
- Compensation (Metabolic acidosis) kidney: PCT: HCO₃ loss
- Acute: For 10 mm Hg ↓ in CO₂: HCO₃ ↓ by 2 mEq
- Chronic: 10 mm Hg ↓ in CO₂: HCO₃ ↓ by 4 mEq

CAUSES

- CNS: Pain, Anxiety, Hysteria, Fever, CVA
- Cardiac Failure : Pulmonary edema
- High-altitude pulmonary edema
- Pulmonary embolism: ↑ Dead Space
- Hemothorax: Chest Trauma
- Miscellaneous: Sepsis, Heat exposure

CLINICAL FEATURES

- ↓ in CO₂: Cerebral hypoperfusion (CO₂ has a **vasodilator** effect on cerebral blood vessels)
 - Dizziness, Vertigo
 - Intracellular shift of sodium, potassium, and phosphate: Arrhythmias can occur
 - Arrhythmia
 - ↑ in the protein binding of Calcium → ↓ in availability of ionized Ca⁺⁺, → causing Tetany



Open lung breathing

Important Information

- M/c acid base disturbance in critically ill patients: **Chronic Respiratory Alkalosis**
- Progesterone increases ventilation and lowers $p\text{CO}_2$ by 5-10 mm Hg, hence pregnancy leads to **Chronic Respiratory alkalosis**

PROCEDURAL ASPECTS

MODIFIED ALLEN'S TEST

- Done to check the integrity of the collateral circulation in superficial palmar arch
- Done before
 - Radial artery puncture or cannulation for sampling
 - Radial artery harvesting before CABG



00:31:40

STEPS

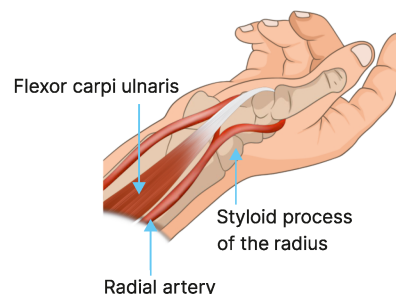
- Ask the patient to form a fist
- Obliterate both Arteries (Ulnar and radial artery)
- Now ask the patient to open palm and notice entire palm is white / pale
- Release the ulnar artery first (keeping radial artery obliterated)
- If Radio ulnar collaterals circulation is sufficient, the whole hand will have a pink healthy flush from the release of either artery

ALTERNATIVE TECHNIQUE TO ASSESS RADIO ULNAR COLLATERAL CIRCULATION

- Color Doppler study of flow
- Plethysmography
- Magnetic Resonance Imaging (MRI)

MUST DO'S WHILE PERFORMING ABG

- Preferred site : radial artery
- Palpate the artery between the flexor carpi radialis tendon and the styloid process
- Press on the area for 5 mins after withdrawing the blood - to prevent hematoma



INSTRUMENTS NEEDED

- 3 ml pre-heparinized syringe - Sample 2ml
- Rubber stopper
- Rolled towel
- Bag of ice

Extra mile:

Excess heparin will causes falsely low pH & low HCO_3^-





Q. A 24-year-old beggar is found unconscious at the gate of Metro Station Green Park. In the ER, room air arterial blood gas is performed and reveals pH 7.25, PCO_2 60mm Hg, PO_2 65mm Hg, HCO_3^- 26meq/L, and Base Excess 1. On the chemistry panel, the sodium is 137meq/L, chloride is 100meq/L, bicarbonate is 27meq/L. What is the diagnosis?

Ans - Uncompensated Respiratory acidosis

Q. A 60-year-old man with amyotrophic lateral sclerosis is brought to the clinic by his family, who are concerned that he is more somnolent than normal. On further history, they report that he has been having morning headaches and does not feel refreshed when he wakes up. An arterial blood gas reveals pH 7.37, PCO_2 57mm Hg, PO_2 70mm Hg, HCO_3^- 34meq/L. What is the diagnosis?

Ans - Fully compensated chronic Respiratory Acidosis

Q. Known case of M. Gravis present with respiratory difficulty. ABG readings shows pH 7.3, PaCO_2 68 mmHg, HCO_3^- 28 mmol/L, and PaO_2 60 mm Hg. How would you interpret this?

Ans - Partially compensated Respiratory Acidosis or Respiratory acidosis with metabolic alkalosis

Q. A 15-year-old juvenile with a history of inhalant abuse presents to the emergency room complaining of dyspnea. He has a SpO_2 of 99% on room air and is tachypneic. A room air arterial blood gas reveals pH 7.25, PCO_2 17mm Hg, PO_2 128 mm Hg, HCO_3^- 2meq/L. A chemistry panel revealed sodium of 130, chloride of 98, and HCO_3^- 2meq/L. What is the diagnosis?

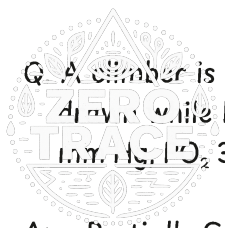
Ans - Partially Compensated Metabolic Acidosis

Q. A three-year-old child is admitted to the hospital with a diagnosis of severe asthma exacerbation. The mother of the child reports to the nurse on duty that she has witnessed slight tremors and behavioral changes in her child over the past four days. The attending physician orders routine ABGs following an assessment of the ABCs. The ABG results are pH 7.35, PaCO_2 72 mm Hg, and HCO_3^- 38 mEq/L. What acid-base disorder is shown?

Ans: Fully Compensated Respiratory Acidosis (Chronic)

Q. A resident working in the ER decides to draw a room air arterial blood gas from a sick patient, which reveals pH 7.46, PCO_2 49 mm Hg, PO_2 68 mm Hg, and HCO_3^- 34meq/L. On his chemistry panel, the sodium is 139, chlorides 95, HCO_3^- 34 meq/L. What is the diagnosis?

Ans: Partially Compensated Metabolic Alkalosis



Q. A climber is coming down from the summit of Mt. Everest. At an altitude of 6000m, he has a blood gas drawn while breathing ambient air as part of a research project. The blood gas reveals pH 7.55, PCO_2 12 mm Hg, PO_2 30 mm Hg and HCO_3^- 20 meq/L. What does it demonstrate?

Ans: Partially Compensated Respiratory Alkalosis

Q. A 57-year-old woman presents with 2 days of fever, dyspnea, and a cough productive of rust-colored sputum. Her room air oxygen saturation in the emergency room is 85%, and the intern decides to obtain a room air arterial blood gas while they are waiting for the CXR to be done. The blood gas reveals: pH 7.54, PCO_2 25 mm Hg, PO_2 65 mm Hg, HCO_3^- 22 meq/L, Base excess -1. What is the diagnosis?

Ans Uncompensated Respiratory Alkalosis

Q. A patient has pH 7.6, PaO_2 120 mm Hg, PaCO_2 31 mm Hg, and HCO_3^- 25 mmol/L. What does this mean?

Ans Uncompensated Respiratory Alkalosis

Q. Patient X is admitted to the hospital and is to undergo brain surgery. He is very anxious and scared of the upcoming surgery. He begins to hyperventilate and becomes very dizzy. STAT ABGs reveal pH 7.51, PaCO_2 22 mm Hg, and HCO_3^- 25 meq/L. What is the ABG interpretation based on the findings?

Ans Uncompensated Respiratory Alkalosis

Q. Hemothorax patient has ABG Lab values: pH 7.42, PaCO_2 25 mm Hg, HCO_3^- 18 meq/L. Interpretation is?

- a. Metabolic acidosis, partially compensated
- b. Respiratory acidosis, partially compensated
- c. Respiratory acidosis, fully compensated
- d. Respiratory alkalosis, fully compensated

Ans (d)

Q. Lab values: pH 7.56, PaCO_2 20 mm Hg, HCO_3^- 20 meq/L?

- a. Respiratory alkalosis, fully compensated
- b. Respiratory alkalosis, uncompensated
- c. Respiratory alkalosis, partially compensated
- d. Metabolic alkalosis, partially compensated

Ans (c)



ABG ANALYSIS SIMPLIFIED PART 2

COMPARISON OF ABG AND VBG VALUES

00:00:13

PARAMETER	ABG (ARTERIAL BLOOD GAS)	VBG (VENOUS BLOOD GAS)	DIFFERENCE
pH	7.35 - 7.45	7.30 - 7.40	~0.03 - 0.05 lower in VBG
pCO ₂ (mmHg)	35 - 45	40 - 50	~5 - 10 mmHg higher in VBG
pO ₂ (mmHg)	80 - 100	30 - 50	Significantly lower in VBG
HCO ₃ ⁻ (mEq/L)	22 - 26	22 - 26	Minimal difference
Base Excess	-2 to +2	-2 to +2	Minimal difference
Lactate (mmol/L)	0.5 - 2.0	0.5 - 2.0	Minimal difference

INDICATIONS WHERE ABG IS A MUST

- VBG can be used as substitute for ABG in cases of metabolic acidosis/general acid-base assessment. ABG is mandatory in following scenarios -
 - Respiratory Failure - To assess oxygenation (PaO₂) and ventilation precisely
 - Severe Hypoxemia - When accurate PaO₂ measurement required
 - ARDS (acute respiratory distress syndrome) - For monitoring oxygenation status (PaO₂/ FiO₂)
 - Mechanical Ventilation - To assess and adjust ventilatory support
 - Severe Hemodynamic Instability - To monitor perfusion and oxygen delivery
 - CO₂ Retention (COPD Exacerbation) - As VBG may overestimate PaCO₂
 - Carbon Monoxide Poisoning - ABG with co-oximetry required to measure carboxyhemoglobin levels

CASE 1

00:02:59

Q. A 47-year-old alcoholic presents with a two-day history of severe abdominal pain and vomiting episodes. On examination, BP is 90/50 mmHg, and he was markedly tender in his epigastrium. His labs reveal Sodium - 132 meq/L, Chloride - 92 meq/L, Creatinine - 1.5 mg/dl, Amylase - 400 U/L and Lipase - 250 U/L. A room air arterial blood gas reveals pH - 7.3, pCO₂ - 31 mm Hg, pO₂ - 88 mm Hg, HCO₃ - 16 meq/L. What is the diagnosis?

Explanation

- Diagnosis: Acute pancreatitis
- ABG report shows:
 - pH = 7.30 → Low → Acidosis

- $p\text{CO}_2 = 31$ mmHg (normal: 35-45) → Low
- $\text{HCO}_3^- = 16$ mEq/L (normal: 22-26) → Low

- Primary change: **Metabolic acidosis**
- To check for compensation: Expected and actual $p\text{CO}_2$ values should be matching
 - Expected $p\text{CO}_2 = \text{HCO}_3^- + 15 = 16 + 15 = 31$ mm Hg
 - Actual $p\text{CO}_2 = 31$ mm Hg
- Anion gap: $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) = 132 - (92 + 16) = 24$ meq/L
- **Final interpretation** - High anion gap metabolic acidosis with compensatory respiratory alkalosis

CASE 2

Q. ABG shows pH - 7.2, $p\text{CO}_2$ - 27 mmHg and HCO_3^- - 12 meq/L. What is the diagnosis?

Explanation

- ABG report shows:
 - pH = 7.20 → Low → Acidosis
 - $p\text{CO}_2 = 27$ mmHg (normal: 35-45) → Low
 - $\text{HCO}_3^- = 12$ mEq/L (normal: 22-26) → Low
- Primary change: **Metabolic acidosis**
- Expected $p\text{CO}_2 = \text{HCO}_3^- + 15 = 12 + 15 = 27$
- Actual $p\text{CO}_2 = 27$
- Interpretation:
 - Partially compensated metabolic acidosis
 - Or
 - Metabolic acidosis with compensatory respiratory alkalosis

CASE 3

00:08:08

Q. Lab values: pH - 7.3, $p\text{CO}_2$ - 32 mmHg, HCO_3^- - 17 meq/L. What is the diagnosis?

- Fully compensated metabolic acidosis
- Partially compensated metabolic acidosis
- Partially compensated respiratory acidosis
- Fully compensated metabolic alkalosis

Ans. (b)

Explanation

- ABG report shows:
 - pH = 7.30 → Low → Acidosis
 - $p\text{CO}_2 = 32$ mmHg → Low
 - $\text{HCO}_3^- = 17$ mEq/L → Low
- Primary change: Metabolic acidosis
- Expected $p\text{CO}_2 = 17 + 15 = 32$ mmHg
- Actual $p\text{CO}_2 = 32$ mmHg
- Final interpretation: **Partially compensated metabolic acidosis**

CASE 4

00:09:00

Q. Lab values: pH = 7.25, $p\text{CO}_2$ = 23 mmHg, HCO_3^- = 8 mEq/L. What is the diagnosis?

- a. Metabolic acidosis with compensatory respiratory alkalosis
- b. Metabolic acidosis with respiratory acidosis
- c. Respiratory acidosis with compensatory metabolic alkalosis
- d. Metabolic acidosis with metabolic alkalosis

Ans. (a)

Explanation

- ABG report shows:
 - pH = 7.25 → Low → Acidosis
 - $p\text{CO}_2$ = 23 mmHg → Low
 - HCO_3^- = 8 mEq/L → Low
- Primary change: Metabolic acidosis
- Expected $p\text{CO}_2$ = $8 + 15 = 23$ mmHg
- Actual $p\text{CO}_2$ = 23 mmHg
- **Final interpretation:** Metabolic acidosis with compensatory respiratory alkalosis

CASE 5

Q. Lab values: pH = 7.30, $p\text{CO}_2$ = 30 mmHg, HCO_3^- = 15 mEq/L. Which of the following will cause this derangement?

- a. Congenital hypertrophic pyloric stenosis
- b. Cushing syndrome
- c. Renal vein thrombosis
- d. Acute exacerbation of chronic bronchitis

Ans. (c)

Explanation:

- Expected $p\text{CO}_2$ = $\text{HCO}_3^- + 15 = 30$ mmHg
- Actual $p\text{CO}_2$ = 30 mmHg
- The values suggest metabolic acidosis
- Congenital hypertrophic pyloric stenosis → cause vomiting → metabolic alkalosis
- Cushing syndrome: cortisol ↑ → ENaC activation → hypokalemic metabolic alkalosis
- Renal vein thrombosis → causes AKI → **Metabolic acidosis**
- Acute exacerbation of chronic bronchitis → causes respiratory acidosis, $p\text{CO}_2$ values will be elevated

CASE 6

00:12:33

Q. 18-month-old baby is brought with episodes of lethargy and multiple vomiting episodes after cough syrup ingestion for URI. Child has not passed urine since last night. On examination baby was obtunded, but she was easily arousable, and muscle tone was normal. The respiratory rate was 30/min. BP was 120/80 mmHg.



Lab work up:

Metabolic profile: Na^+ - 130 meq/l, K^+ - 4.2 mEq/l, BUN and creatinine are elevated

Urinalysis: pH - 5.0; negative for glucose & ketones. Calcium oxalate crystals were seen on urine microscopy

Arterial Blood Gas report: pH - 7.19, pCO_2 - 16 mmHg, PO_2 - 100 mmHg, HCO_3^- - 6.2 mEq/l

Diagnosis?

- Euvolemic hyponatremia
- Acute tubular necrosis
- Renal tubular acidosis
- Sepsis

Ans. (b)

Explanation

- Urinalysis Findings:
 - Urine pH = 5.0 (acidic)
 - No glucose/ketones
 - Calcium oxalate crystals → Strongly suggest **ethylene glycol poisoning**, especially in a child after cough syrup ingestion
 - Ethylene glycol poisoning cause **ATN** (Acute tubular necrosis)
- ABG Interpretation: **Metabolic acidosis**
 - pH = 7.19 → Acidotic
 - pCO_2 = 16 mmHg (↓)
 - HCO_3^- = 6.2 mEq/L (↓↓)
- Euvolemic hyponatremia seen with SIADH
- Renal tubular acidosis: Urine output will be normal, but here it is reduced
- Sepsis → result in septic shock → Hypotension

CASE 7

00:18:04

Q. 35-year-old man underwent emergent abdominal surgery and has a nasogastric tube in situ. The nurse on duty notes that the nasogastric tube is draining a large amount (900 cc in 2 hours) of coffee ground secretions. The patient is not oriented to person, place, or time, so STAT ABG is ordered by you. The results from the ABG shows pH - 7.57, PaCO_2 - 45 mmHg, and HCO_3^- - 30 mEq/L. What is your assessment?

- Respiratory acidosis, partially compensated
- Respiratory alkalosis, partially compensated
- Metabolic acidosis, partially compensated
- Metabolic alkalosis, partially compensated

Ans. (d)

Explanation:

- Nasogastric tube is draining a large amount of coffee ground secretions indicate **Erosive Gastritis** contributing to GI bleed



- ABG
 - pH: 7.47 → alkalotic
 - PaCO_2 : 45 → normal
 - HCO_3^- : 30 mEq/L → high
- Interpretation: Metabolic alkalosis
- Expected $\text{pCO}_2 = 30 + 15 = 45$ mmHg
- Actual $\text{pCO}_2 = 45$ mmHg
- **Final interpretation:** Metabolic alkalosis with partial respiratory compensation

CASE 8

00:21:07

- Q. Lab values: pH - 7.52, pCO_2 - 48 mm Hg, HCO_3^- - 33 meq/L
- Respiratory alkalosis, partially compensated
 - Respiratory alkalosis, partially compensated
 - Metabolic acidosis, partially compensated
 - Metabolic alkalosis, partially compensated

Ans. (d)

Explanation

- ABG report:
 - pH = 7.52 → High → Alkalosis
 - $\text{pCO}_2 = 48$ mmHg → Slightly high
 - $\text{HCO}_3^- = 33$ mEq/L → High
- Primary change: Metabolic alkalosis
- To check for compensation:
 - Expected $\text{pCO}_2 = 33 + 15 = 48$ mmHg
 - Actual $\text{pCO}_2 = 48$ mmHg
- **Final interpretation:** Metabolic alkalosis, partially compensated / Metabolic alkalosis with compensatory respiratory acidosis

CASE 9

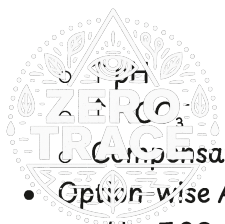
00:22:21

- Q. You are reviewing the latest arterial blood gas results for a sick Cushing's Syndrome patient. Which of the following result indicates that the ABG derangement is compensated?
- pH - 7.32
 - PaCO_2 - 18 mmHg
 - HCO_3^- - 8 mEq/L
 - PaCO_2 - 48 mmHg

Ans. (d)

Explanation:

- Cushing syndrome:
 - Cortisol ↑ → ENaC activation → ↑ Na^+ reabsorption, ↑ K^+/H^+ loss → metabolic alkalosis
- In metabolic alkalosis, we expect:



Compensatory $\uparrow$ PaCO₂ (respiratory compensation)

- Option-wise Analysis:
 - pH = 7.32 → Acidosis, not consistent with metabolic alkalosis
 - PaCO₂ = 18 mmHg → Very low → suggests respiratory alkalosis
 - HCO₃⁻ = 8 mEq/L → Very low → metabolic acidosis
 - PaCO₂ = 48 mmHg → Elevated, suggesting **respiratory compensation for metabolic alkalosis**

CASE 10

- Q. A patient is post-op from knee surgery. The patient has been receiving analgesia I/V every 2 hours. You notice the patient is exhibiting a respiratory rate of 8/min and is extremely drowsy. Which of the following conditions is the patient at risk of?
- a. Respiratory acidosis
 - b. Respiratory alkalosis
 - c. Metabolic alkalosis
 - d. Metabolic acidosis

Ans. (a)

Explanation:

- High dose opioids → development of respiratory depression
- When RR is reduced → $\uparrow$ CO₂ in the body → **Respiratory acidosis**

Important Information

- In case of inadequate analgesia → excessive pain → $\uparrow$ RR → **Respiratory alkalosis**

CASE 11

- Q. A patient attempted to commit suicide by ingesting a bottle of Aspirin. Which of the following conditions is this patient at risk for?
- a. Hyperkalemia
 - b. Hypercalcemia
 - c. Respiratory alkalosis
 - d. Respiratory acidosis

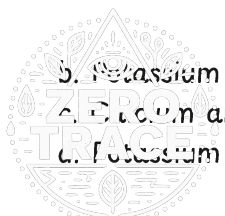
Ans. (c)

Explanation:

- Aspirin: Acetyl salicylic acid → stimulate respiratory centre → CO₂ washout → **Respiratory alkalosis**

CASE 12

- Q. Respiratory alkalosis patient is having circumoral muscle twitching and muscle weakness. Which of the following electrolyte level derangements is most likely to be responsible for this presentation?
- a. Calcium and Sodium levels



- b. Potassium and Sodium levels
- a. Calcium and Potassium levels
- c. Potassium and Phosphate levels

Ans. (c)

Explanation:

- Respiratory alkalosis → redistribution of Calcium
 - ↓ Ionized calcium, bound calcium to albumin ↑
 - Intracellular shift of Na^+ , K^+ , PO_4^-
 - ↓ K^+ → muscle cramps, muscle weakness
- In the given scenario:
 - Circumoral muscle twitching d/t low calcium
 - Muscle weakness d/t low K^+

CASE 13

Q. A Patient is experiencing respiratory alkalosis. What is the most classic sign seen?

- a. Bradypnea
- b. Chvostek sign
- c. Bradycardia
- d. Hoover sign

Ans. (b)

Explanation:

- Classic sign will be of Tetany → Chvostek sign
- Hoover's sign - differentiate hysterical (functional) weakness from an organic (neurological) cause

CASE 14

Q. A patient has the following blood gases: pCO_2 - 25 mm Hg, pH - 7.50, HCO_3^- - 19 meq/L. Which of the following could not be the cause of this condition?

- a. Anxiety attack
- b. pO_2/FiO_2 ratio <100
- c. Hysteria
- d. Aspirin toxicity

Ans. (b)

Explanation:

- Anxiety attack ↑ Co_2 washout d/t ↑ RR
- pO_2/FiO_2 ratio <100 - indicate Severe ARDS ↑ Co_2 build up → Respiratory acidosis
- Hysteria → Cause fast breathing → ↓ Co_2
- Aspirin toxicity → Stimulate respiratory centre → Hyperventilation → ↓ Co_2

CASE 15

Q. A patient on mechanical ventilation has the following blood gases: PaCO_2 - 29 mm Hg, pH - 7.56, HCO_3^- - 23 mEq/L. Which of the following is primary change in this report?

- a. Respiratory alkalosis
- b. Respiratory acidosis
- c. Metabolic acidosis
- d. Metabolic alkalosis

Ans. (a)

Explanation:

- ABG report shows:
 - pH = 7.56 → High → Alkalosis
 - pCO_2 = 29 mmHg (normal: 35-45) → Low
 - HCO_3^- = 23 mEq/L (normal: 22-26) → Normal
 - Since HCO_3^- is within normal range → Uncompensated
- Elevated pH with low pCO_2 and normal HCO_3^- suggests a primary respiratory cause
- Interpretation: Respiratory alkalosis

CASE 16

Q. A patient is experiencing respiratory acidosis due to brain trauma. Which of the following lab values correlates with this acid imbalance?

- a. Potassium level of 6.0
- b. Potassium level of 2.5
- c. Potassium level of 5.0
- d. Potassium level of 3.5

Ans. (a)

Explanation:

- In Respiratory alkalosis
 - ↓ CO_2
 - Influx of Na^+ , K^+ , PO_4^- → ↓ values of K^+
- In Respiratory acidosis
 - CO_2 Retention → acid build up in blood
 - To buffer this acidosis → H^+ moves inside and K^+ moves out into extracellular space → Hyperkalemia
- Among the given values, 6.0 mEq/L reflects hyperkalemia

CASE 17

00:34:06

Q. Which of these patients is experiencing partially compensated respiratory acidosis?

- a. PaCO_2 - 30, pH - 7.35, HCO_3^- - 26
- b. PaCO_2 - 53, pH - 7.23, HCO_3^- - 28
- c. PaCO_2 - 45, pH - 7.49, HCO_3^- - 21
- d. PaCO_2 - 50, pH - 7.30, HCO_3^- - 23

Ans. (b)

Explanation:

- Option A: PaCO_2 - 30, pH - 7.35, HCO_3^- - 26 - can be ruled out
- For respiratory acidosis: PaCO_2 should be $>45\text{mmHg}$
- Option C: PaCO_2 - 45, pH - 7.49, HCO_3^- - 21 - can be ruled out
 - pH is 7.49 $\rightarrow$ alkalotic
- Option D: PaCO_2 - 50, pH - 7.30, HCO_3^- - 23 - can be ruled out
 - HCO_3^- is normal - no compensation
 - Normal value of HCO_3^- : 22-26 mmHg
- Option B: PaCO_2 - 53, pH - 7.23, HCO_3^- - 28
 - HCO_3^- begins to $\uparrow$ - **Partially compensated**

CASE 18

Q. Which of the following is not a cause of respiratory acidosis?

- Pulmonary emboli
- Life threatening asthma
- Chronic obstructive pulmonary disease exacerbation
- Hyperventilation

Ans. (d)

Explanation:

- Life threatening asthma- there will be $\uparrow \text{CO}_2 \rightarrow$ Can be ruled out
- Chronic obstructive pulmonary disease exacerbation $\rightarrow$ there will be $\uparrow \text{CO}_2 \rightarrow$ Can be ruled out
- Hyperventilation $\rightarrow \text{CO}_2$ washout $\rightarrow$ **Respiratory alkalosis**
- Pulmonary embolism:
 - Initially there will be respiratory alkalosis
 - If T/t is delayed $\rightarrow$ respiratory muscle fatigue CO_2 build up $\rightarrow$ Respiratory acidosis

CASE 19

00:38:10

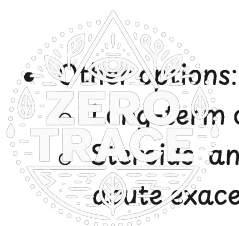
Q. A patient with COPD is admitted due to low spO_2 of 85% in room air with breathing trouble. He has the following blood gases: pCO_2 - 60, pH - 7.20, and HCO_3^- - 34 meq/L. Which of the following is correct about immediate management of this patient?

- Bi-PAP
- Long term oxygen therapy using oxygen concentrator at 1 L min
- I/V hydrocortisone with LABA nebulization
- LABA with LAMA nebulization

Ans. (a)

Explanation:

- ABG interpretation:
 - $\uparrow \text{pCO}_2$ and $\downarrow \text{pH} \rightarrow$ chronic respiratory acidosis
 - $\uparrow \text{HCO}_3^-$ suggests partial renal compensation, typical in chronic CO_2 retention (as seen in COPD)
- Immediate management: Bi-PAP to support ventilation and $\downarrow \text{CO}_2$ levels



• Other options:

• Long-term oxygen therapy- for stable home use

• Steroids and bronchodilators (LABA/LAMA) are adjuncts, not immediate life-saving intervention in acute exacerbation of chronic bronchitis

CASE 20

Q. A 28-year-old woman was admitted electively to a HDU (high dependency unit) following a cesarean section. A diagnosis of 'fatty liver of pregnancy' had been made preoperatively. She was commenced on a continuous morphine infusion at 5 mg/hr and received oxygen by mask. She was noted to be extremely drowsy the next day. Arterial blood gases show pH - 7.16, pCO₂ - 50 mmHg, pO₂ - 115 mmHg, HCO₃⁻ - 15 mEq/L interpret the report?

Explanation:

- ABG Results -
 - pH: 7.16 → Acidemia
 - pCO₂ : 50 mmHg → ↑ (normal: 35-45)
 - HCO₃⁻ : 15 mEq/L → ↓ (normal: 22-26)
- Primary change: Respiratory acidosis
- Expected HCO₃:
 - For every 10mmHg ↑ in CO₂, HCO₃⁻ ↑ by 1 mEq
 - In the given scenario this rule is not satisfied as HCO₃ values are low
- Interpretation: **Respiratory acidosis + Metabolic acidosis**

CASE 21

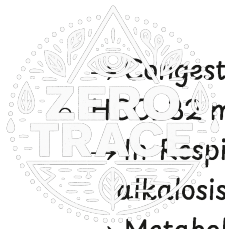
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Q. A 70-year-old man was admitted due to SOB and diagnosed as having acute decompensated cardiac failure. He is put on non-rebreather mask and started on Lasix. On Admission lab reports are shown below Interpret the report?

BIOCHEMISTRY RESULTS	ARTERIAL BLOOD GASES
• Na⁺ - 127 mEq/L • K⁺ - 5.2 mEq/L • Cl⁻ - 79 mEq/L • Urea - 50.5 mg/dl • Creatinine - 0.38 mg/dl • Random blood glucose - 200 mg/dl	• pH - 7.58 • pCO₂ - 21 mmHg • pO₂ - 154 mmHg • HCO₃ - 32 mmol/l

Explanation:

- Biochemistry report analysis
 - Na⁺ - 127 meq/L → low, d/t volume expansion in HF or Lasix induced loss
 - K⁺ - 5.2 meq/L → normal
 - Urea - 50.5 mg/dl ↑, suggesting ↓ renal perfusion d/t HF
- ABG interpretation
 - ↑pH (7.58) and ↓pCO₂ (21 mmHg) → Respiratory alkalosis



→ Congestive HF → Pulmonary edema → Hyperventilation leading to CO_2 washout

HCO_3^- 32 mEq/L ↑ → Metabolic alkalosis

→ In Respiratory alkalosis HCO_3^- should be reducing, but in this case, values are elevated → metabolic alkalosis

→ Metabolic alkalosis might be d/t diuretics (Lasix)

- Interpretation: Respiratory alkalosis with metabolic alkalosis



ABG ANALYSIS SIMPLIFIED PART - 3

CASE - 1

Comment on the ABG report

- pH - 7.30
- $p\text{CO}_2$ - 30 mmHg
- HCO_3 - 10 mEq/L

INFERENCE

- $\downarrow$ pH, $\downarrow$ $p\text{CO}_2$, $\downarrow$ HCO_3
- Primary change - Metabolic acidosis
- Expected $p\text{CO}_2$ - $\text{HCO}_3 + 15 \rightarrow 25$ mmHg
- Actual $p\text{CO}_2$ - 30 mmHg

- Metabolic acidosis with hypoventilation and respiratory acidosis
- Metabolic acidosis with hyperventilation and respiratory alkalosis
- Respiratory acidosis with metabolic alkalosis
- Respiratory alkalosis with metabolic alkalosis

Ans: (a)

- Metabolic acidosis with hyperventilation and respiratory alkalosis if expected and actual $p\text{CO}_2$ are equal
- When actual $p\text{CO}_2 >$ expected $p\text{CO}_2$ = mixed disorder

CASE - 2

A 40-year-old man presents with flash pulmonary edema. Blood work shows:

Random blood Glucose= 243 mg/dL

Blood Urea= 81 mg/dL

Creatinine= 1.5 U/L

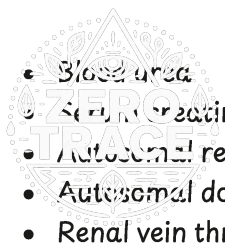
ABG shows respiratory acidosis with metabolic acidosis. Which of the following is a possible cause of this presentation?

- Autosomal recessive polycystic kidneys
- The autosomal dominant tubulointerstitial disease of the kidney
- Renal vein thrombosis
- Renal artery stenosis

Ans: (d)

INFERENCE

- Flash pulmonary edema
- RBG - Diabetes mellitus



- Blood urea } Kidney could be damaged
- Serum creatinine }
- Autosomal recessive polycystic kidneys → ESRD → metabolic acidosis
- Autosomal dominant tubulointerstitial disease of the kidney → ESRD → metabolic acidosis
- Renal vein thrombosis → AKI → metabolic acidosis
- Renal artery stenosis → severe HTN → Left ventricular failure → Cardiac Output ↓ → ↓perfusion → metabolic acidosis
- Two reasons for metabolic acidosis
 - ↓ perfusion
 - Damaged kidneys → unable to excrete H⁺
- Left ventricular failure → pulmonary edema → ↑CO₂ → respiratory acidosis
- Flash pulmonary edema - feature of Renal artery stenosis

CASE - 3

CKD patient with T2DM presents with vomiting. He is on Empagliflozin, Thiazides, Ramipril and bed-time glargine ABG is done. What is the diagnosis?

pH= 7.4

PCO₂ =40 mm Hg

HCO₃ =22 mEq/L

Sodium=140 mEq/L

Chloride=90 mEq/L

Potassium=3.0 mEq/L

- HAGMA with metabolic alkalosis
- NAGMA with metabolic alkalosis
- Normal report
- High anion gap metabolic alkalosis

Ans: (a)

INFERENCE

- Normal pH, pCO₂ and HCO₃
- Anion gap: $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) \rightarrow 140 - 112 \rightarrow 28$ (↑ anion gap)
- Delta Ratio = $(\text{AG} - 12) / (24 - \text{HCO}_3^-)$
= $(28 - 12) / (24 - 22)$
= 8

DELTA RATIO	INTERPRETATION
< 4.0	Pure NAGMA
0.4 - 0.8	HAGMA + NAGMA
0.8 - 2	Pure HAGMA
>2	HAGMA + Metabolic alkalosis

- CKD - metabolic acidosis (kidneys unable to excrete H^+)
- Vomiting - d/t **uremia** can cause metabolic alkalosis
- This patient can develop **metabolic alkalosis**
- This patient has mixed disorder $\rightarrow$ pH, pCO_2 and HCO_3 normal

CASE - 4

Which of the following will differentiate acute from chronic respiratory acidosis in assessing patient with motor neuron disease?

- Increased pCO_2
- Decreased pO_2
- Increased HCO_3
- Decreased HCO_3

Ans: (c)

INFERENCE

- Motor neuron disease has diaphragmatic paralysis $\rightarrow CO_2 \uparrow \rightarrow HCO_3 \uparrow$

CASE - 5

	pH	pCO_2	HCO_3	
1	7.5	51	36	A. Fully compensated respiratory alkalosis
2	7.25	31	16	B. Fully compensated respiratory acidosis
3	7.4	25	18	C. Partially compensated metabolic acidosis
4	7.4	60	34	D. Metabolic alkalosis with respiratory compensation

- 1-D, 2-C, 3-A, 4-B
- 1-C, 2-D, 3-A, 4-C
- 1-A, 2-B, 3-D, 4-C
- 1-B, 2-A, 3-C, 4-D

Ans: (a)

INFERENCE

- Option 1 $\rightarrow$ pH $\uparrow$, $pCO_2 \uparrow$ and $HCO_3 \uparrow \rightarrow$ metabolic alkalosis
- Option 2 $\rightarrow$ pH $\downarrow$, $pCO_2 \downarrow$ and $HCO_3 \downarrow \rightarrow$ metabolic acidosis
- Option 3 $\rightarrow$ pH - normal, $pCO_2 \downarrow$ and $HCO_3 \downarrow \rightarrow$ fully compensated respiratory alkalosis

CASE - 6

Which of the following is the most plausible explanation for the ABG report?

pH = 7.5

$\text{HCO}_3^- = 30 \text{ mEq/L}$

$\text{pCO}_2 = 40 \text{ mm Hg}$

$\text{Na} = 130 \text{ meq/L}$

Chloride = 90 meq/L

- a. Weakening of respiratory muscles effort due to neuromuscular disease
- b. Methanol intoxication
- c. Panic attack in young woman
- d. Vomiting

Ans: (d)

INFERENCE

- (+) alkalosis
- Elevation in $\text{HCO}_3^- \rightarrow$ metabolic alkalosis
- Anion gap: $130 - (90 + 30) \rightarrow 10 \text{ meq/L}$
- Neuromuscular disease will cause respiratory acidosis d/t the involvement of diaphragm
- Methanol intoxication $\rightarrow$ metabolic acidosis
- Panic attack $\rightarrow$ respiratory alkalosis d/t CO_2 washout (fast breathing)
- Vomiting always cause metabolic alkalosis

CASE - 7

Analyze the following report and choose the most likely diagnosis based on the same:

$\text{Na} = 135 \text{ mEq/L}$

$\text{K} = 1.7 \text{ mEq/L}$

$\text{HCO}_3^- = 9 \text{ mEq/L}$

Chloride = 116 mEq/L

$\text{pO}_2 = 80 \text{ mm Hg}$

$\text{pCO}_2 = 24 \text{ mm Hg}$

$\text{pH} = 7.25$

- a. Non-anion gap metabolic acidosis
- b. Respiratory acidosis with compensatory metabolic alkalosis
- c. High anion gap metabolic acidosis
- d. Metabolic alkalosis

Ans: (a)

INFERENCE

- $\text{pH} \downarrow, \text{pCO}_2 \downarrow, \text{HCO}_3^- \downarrow$
- Metabolic acidosis